

DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

Contract No.: DM001		CDRL No.:	
Project Title: DUBAI METRO		CDRL Title:	
Document Title: Red Line – Rashidiya Main Depot Building 6 Diesel Engine Workshop Structural Design Calculation (AHU Support Platform) DCP 4 Submission			
Revision History			
A1	18-Oct-08	First Issue	Signed below
MARK	DATE	DESCRIPTION	CIVIL
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DATE	-	DATE	18 October 2008
RAIL SYSTEM		CIVIL JV	
CONTRACTOR'S DOCUMENT No.:		DOCUMENT No.:	REVISION
		DM001-E-MAG-CDP-DR-DCC-371921	A1

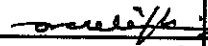
**Dubai Metro, UAE
Rashidiya Main Depot**

**Structural Design Calculation
(AHU Support Platform)
Building 6 – Diesel Engine Workshop**

October 2008

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-MAG-CDP-DR-DCC-371921	Controlled Copy No.:
Document Title: Main Depot- Rashidiya Structural Design Calculation (AHU Support Platform) Building 6 – Diesel Engine Worksop	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 8 Oct 2008

Revision, Review and Approval/Authorization Records

A1	Issued for Approval	 CK Lam Oct 08	 Drake Lawale Oct 08	 B. Maney Oct 08
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

Controlled Copy No.	Issued to
1	JT Metro JV
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4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Document No.: DM001-E-MAG-CDP-DR-DCC-371921

Rev A1

Date: 8 Oct 2008
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1 INTRODUCTION

1 INTRODUCTION

Due to the changes of enhanced air conditioning requirements in Depot buildings and increase in weight of AHU equipments, this supplementary calculation supports the proposal of mounting the AHUs on raised platform that spans across the tracks.

This is outlined in JTMJV's Instruction Ref JTM/AT/DM001/CIC/08-272 dated 26 June 2008.

The design consideration is based on the followings:

- The necessary investigation/evaluation on the AHU structural support methods and AHU installation/maintenance system.
- Latest AHU equipment information.
- Site constraints.
- Feedback from various party (JTMJV, MHI, M&E, etc) and agreements made in joint coordination meetings.

The proposed structural system is to use a column supporting steel platform floors to take all the gravity loads (including the AHU equipment units). The steel frames are designed with pinned connection at support location, and the integrated cross bracing system (consists of the either vertical cross bracing and/or angle knee bracing) gives the stability for the structure in resisting the lateral forces. In addition, the steel plate at platform level will give additional restraints to the frames. The proposed structural frame with bracing system will be able to simplify the connections between the beams and columns and increase the efficiency of the columns.

Supplementary calculations are appended in Appendix A of this report, and the layout plans are enclosed in this report for reference as well (see Appendix B).

Appendix A

Supplementary Design Calculations

Appendix A-1

Platform Structural Members Design

Site

Ras B6

Date

Job No.

AHU Platform

Designed by OEH

Sheet No.

Loadings :

DL :

Beam s/w

1 kN/m (Assumed)

Floor s/w
(6mm thk plate) 0.006×78.5 $= 0.48 \text{ kN/m}^2$ $= 1 \text{ kN/m}^2$ (Assume 2m load width)

M&E services

0.5 kN/m²

LL :

LL (Maintenance)

1.5 kN/m²

AHU

30 kN/unit

Site RAS B6

Date

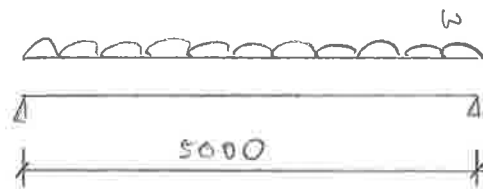
Job No.

AHU platform

Designed by OEH

Sheet No.

Beam size selected: B3



(without AHU load)

Loading width = 0.9 m

$$W_{DL} = 1 + 1 + 0.5 \times 0.9 = 2.5 \text{ kN/m}$$

$$W_{LL} = 1.5 \times 0.9 = 1.4 \text{ kN/m}$$

At ULS:

$$W = 1.4 \times 2.5 + 1.6 \times 1.4 = 5.8 \text{ kN/m}$$

$$M_{max} = \frac{1}{8} \times 5.8 \times 5^2$$

$$= 19 \text{ kNm}$$

$$V_{max} = \frac{1}{2} \times 5.8 \times 5$$

$$= 15 \text{ kN}$$

Site RAJ B6

Date

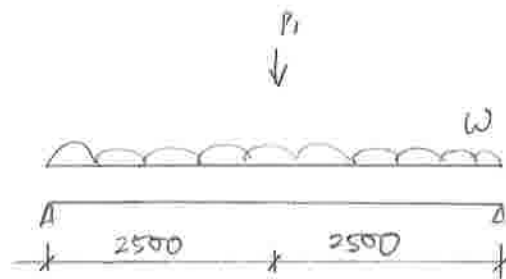
Job No.

AHU Platform

Designed by OEM

Sheet No.

Beam size selected = B3



(with AHU load)

Loading width = 0.9 m

$$W_{DL} = 1 + 1 + 0.5 \times 0.9 = 2.5 \text{ kN/m}$$

$$W_{LL} = 1.5 \times 0.9 = 1.4 \text{ kN/m}$$

$$P_{i,LL} (\text{AHU}) = 30/2 = 15 \text{ kN}$$

At ULS:

$$W = 1.4 \times 2.5 + 1.6 \times 1.4 = 5.8 \text{ kN/m}$$

$$P_1 = 1.6 \times 15 = 24 \text{ kN}$$

$$\begin{aligned} M_{\max} &= \frac{1}{8} \times 5.8 \times 5^2 + \frac{1}{4} \times 24 \times 5 \\ &= 19 + 30 \\ &= 49 \text{ kNm} \end{aligned}$$

If P_1 is at near support

$$\begin{aligned} V_{\max} &= \frac{1}{2} \times 5.8 \times 5 + 24 \\ &= 39 \text{ kN} \end{aligned}$$

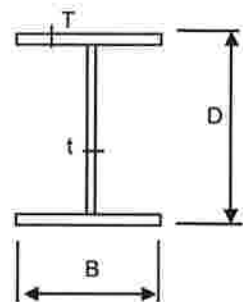
- Steel Beam**
- Check for M_b
 - UB with equal flange
 - For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties

Build-up Section

Simply-supported beam

Section	=	w	("r" for Rolled section, "w" for Welded section)	
p_y	=	345	N/mm2	
E	=	205000	N/mm2	
D	=	300	mm	} $b/T = 9.7 < 13\varepsilon$ $d/t = 46.7 < 80\varepsilon$ $\Rightarrow$ Class 3 semi-compact section
B	=	200	mm	
t	=	6	mm	
T	=	10	mm	
				B3 (Beam / Column Mark)
A	=	56.8	cm2	= 5680 mm2
I_x	=	9510.93	cm4	= 95109333.33 mm4
I_y	=	1333.87	cm4	= 13338733.33 mm4
Z_x	=	634.06	cm3	= 634062 mm3
Z_y	=	133.39	cm3	= 133387 mm3
S_x	=	697.60	cm3	= 697600 mm3
S_y	=	33.35	cm3	= 33346.83333 mm3
r_x	=	12.94	cm	= 129.400931 mm
r_y	=	4.85	cm	= 48.45996848 mm



Design Data

M_x	=	49	kNm	
V	=	39	kN	
L	=	5	m	Overall span
L_e	=	5	m	Effective length for M_b

Design for Bending Moment

λ	=	103	
From BS5950 clause. 4.3.6.7			
u	=	1.00	
x	=	30.00	
λ/x	=	3.44	
v	=	0.89	
Sect'n class	=	3.00	Equal flange { 1 for class 1 plastic 2 for class 2 compact 3 for class 3 semi-compact 4 for slender
β_w	=	0.9	

$$\lambda_{LT} = uv\lambda(\beta_w)^{0.5} = 88$$

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	161	N/mm ²
M_b	=	102.1	kNm

From Table 18

m_{LT}	=	1	
M_b/m_{LT}	=	102.1	kNm
	>	M_x	= 49.0 kNm OK !

Design for Bending Shear

A_v	=	1680	mm ²
P_v	=	348	kN
$0.6P_v$	=	209	kN
	>	V	= 39.0 kN Low shear, OK !

Site Res B6

Date

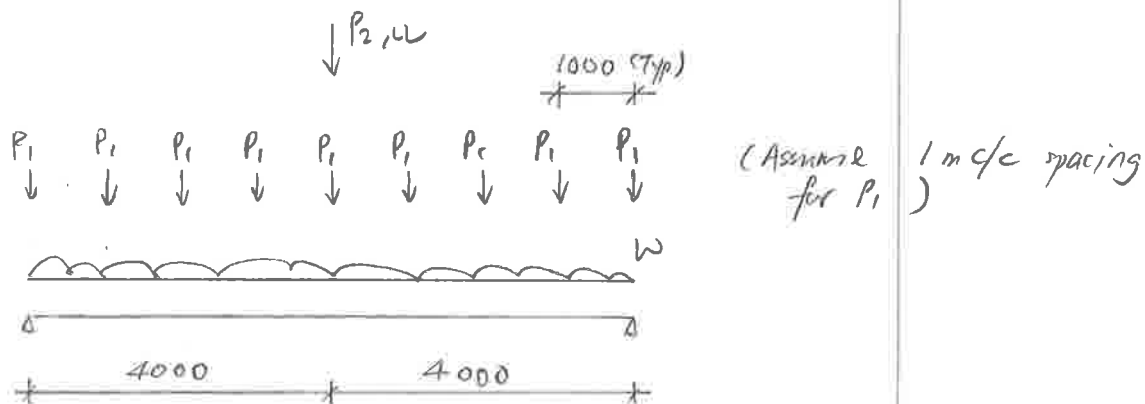
Job No.

AHU Platform

Designed by OEH

Sheet No.

Beam size selected = B23



$$w_{DL} = 1.5 \text{ kN/m (Beam s/w)}$$

$$P_{2,UL} = 30/2 = 15 \text{ kN}$$

At ULS

$$w = 1.4 \times 1.5 = 2.1 \text{ kN/m}$$

$$P_1 = 15 \text{ kN} \times 1.2 = 17 \text{ kN} \equiv w_1 = 17 \text{ kN/m}$$

$$P_{2,UL} = 1.6 \times 15 = 24 \text{ kN}$$

$$M_{max} = \frac{1}{8} (2.1 + 17) \times 8^2 + \frac{1}{4} \times 24 \times 8$$

$$= 153 + 48$$

$$= 201 \text{ kNm}$$

If $P_{2,UL}$ is near support

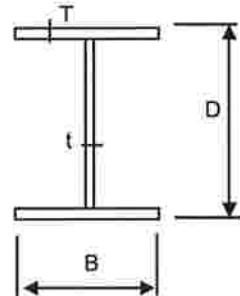
$$V_{max} = \frac{1}{2} (2.1 + 17) \times 8 + 24$$

$$= 101 \text{ kN}$$

- Steel Beam**
- Check for M_b
 - UB with equal flange
 - For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties **Build-up Section** **Simply-supported beam**

Section	=	w	("r" for Rolled section, "w" for Welded section)	
p_y	=	345	N/mm2	
E	=	205000	N/mm2	
D	=	500	mm	$b/T = 8.0 < 9\epsilon$ $d/t = 47.0 < 80\epsilon$ ⇒ Class 2 compact section
B	=	250	mm	
t	=	10	mm	
T	=	15	mm	
B23 (Beam / Column Mark)				
A	=	122	cm2	= 12200 mm2
I_x	=	52770.67	cm4	= 527706666.7 mm4
I_y	=	3910.42	cm4	= 39104166.67 mm4
Z_x	=	2110.83	cm3	= 2110827 mm3
Z_y	=	312.83	cm3	= 312833 mm3
S_x	=	2371.00	cm3	= 2371000 mm3
S_y	=	78.21	cm3	= 78208.33333 mm3
r_x	=	20.80	cm	= 207.9775103 mm
r_y	=	5.66	cm	= 56.61501182 mm



Design Data

M_x	=	201	kNm	
V	=	101	kN	
L	=	8	m	Overall span
L_e	=	8	m	Effective length for M_b

Design for Bending Moment

λ	=	141	
From BS5950 clause. 4.3.6.7			
u	=	1.00	
x	=	33.33	
λ/x	=	4.24	
v	=	0.85	
Sect'n class	=	2.00	
β_w	=	1.0	
Equal flange { 1 for class 1 plastic 2 for class 2 compact 3 for class 3 semi-compact 4 for slender			
λ_{LT}	=	$uv\lambda(\beta_w)^{0.5}$	= 120

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	103	N/mm ²
M_b	=	244.2	kNm

From Table 18

m_{LT}	=	1	
M_b/m_{LT}	=	244.2	kNm
	>	M_x	= 201.0 kNm OK !

Design for Bending Shear

A_v	=	4700	mm ²
P_v	=	973	kN
$0.6P_v$	=	584	kN
	>	V	= 101.0 kN Low shear, OK !

Appendix A-2

Check on Existing Foundation

Site Ras B6

Date

Job No.

AHU platform

Designed by OEH

Sheet No.

Hanger column loading (Tension)

AHU platform with 1 no. of AHU unit

$$\text{Loaded area} \approx 2.5 \times 4 = 10 \text{ m}^2$$

DL:

$$W_{DL} = 0.5 + 0.5 = 1 \text{ kN/m}^2$$

$$\begin{aligned} \text{Beam sw} &= 2.5 \times 5 \text{ nos.} \times 0.5 \text{ kN/m} + 4 \times 1 \text{ no.} \times 1.0 \text{ kN/m} \\ &= 10.3 \text{ kN} \end{aligned}$$

$$N_{DL} = 1 \times 10 + 10.3 = 20.3 \text{ kN}$$

LL:

$$W_{LL} = 1.5 \text{ kN/m}^2$$

$$P_{LL, AHU} = 30 \text{ kN}$$

$$N_{LL} = 1.5 \times 10 + 30 \times 3/4 \times 1/2 = 26.3 \text{ kN}$$

Total column load

$$N_{SL} = 46.6 \text{ kN}$$

$$N_{UL} = 1.4 \times 20.3 + 1.6 \times 26.3 = 70.5 \text{ kN}$$

Main Frame

Column additional loading (Through 2 supports)

$$N_{SL} = 46.6 \times 2 = 93.2 \text{ kN}$$

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site Main Depot Rashidiya - Building 6
Footing Summary

Date Oct '08 Job No. 068010

Designed by OEH Sheet No.

Allowable soil Bearing Pressure = 150 kN/m²

Soil Density = 20 kN/m³

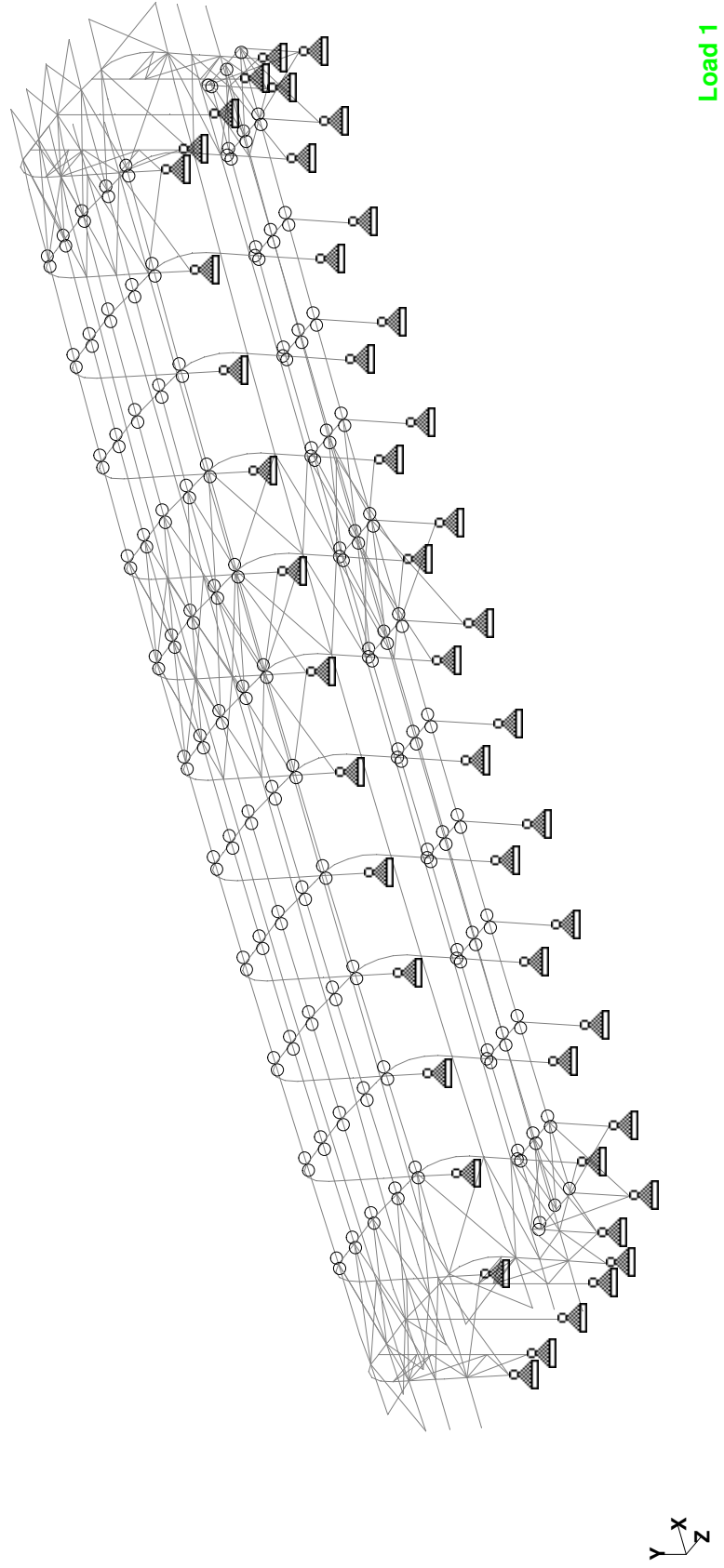
Concrete Density = 24 kN/m³

GL	Footing Type	Footing Length (m)	Footing Width (m)	Footing Depth (m)	STAAD Column ref	STAAD Column Reaction (ULS) (kN)*	Max Vertical Crane Load Combination (kN)	AHU Platform Load (kN)	Footing Weight (kN)	Soil Depth (m)	Soil Weight (kN)	Total Load (kN)	Bearing Pressure (kN/m ²)	Remark
2/A	F25	3.30	1.80	0.750	1301	436	237	93	107	1.20	143	891	150	Ok
2/B	F25	3.30	1.80	0.750	1325	458	237	93	107	1.20	143	907	153	Acceptable
3/A	F25	3.30	1.80	0.750	1401	461	237	93	107	1.20	143	909	153	Acceptable
3/B	F25	3.30	1.80	0.750	1425	422	237	93	107	1.20	143	881	148	Ok
11/A	F25	3.30	1.80	0.750	2201	474	237	93	107	1.20	143	918	155	Acceptable
11/B	F25	3.30	1.80	0.750	2225	527	237	93	107	1.20	143	956	161	Acceptable
12/A	F25	3.30	1.80	0.750	2301	434	237	93	107	1.20	143	890	150	Ok
12/B	F25	3.30	1.80	0.750	2325	451	237	93	107	1.20	143	901	152	Acceptable

Appendix A-3

Check on Main Frame Steel Members

STAAD Input File – 3D Frame With AHU Platform Loading



Whole Structure

STAAD SPACE
START JOB INFORMATION
ENGINEER DATE April 2007
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN

JOINT COORDINATES

*MAIN HALL

1002	-52.000	3.200	-8.049;	1004	-52.000	5.600	-8.299;
1005	-52.000	7.419	-8.488;	1010	-52.000	11.585	-6.210;
1012	-52.000	11.756	-3.400;	1013	-52.000	11.885	0.000;
1014	-52.000	11.756	3.400;	1016	-52.000	11.585	6.210;
1021	-52.000	7.419	8.488;	1022	-52.000	5.600	8.299;
1024	-52.000	3.200	8.049;				
1101	-48.000	0.000	-7.715;	1102	-48.000	3.200	-8.051;
1103	-48.000	4.926	-8.233;	1104	-48.000	5.600	-8.304;
1105	-48.000	7.600	-8.514;	1106	-48.000	8.800	-8.640;
1107	-48.000	9.849	-8.543;	1108	-48.000	10.781	-8.055;
1109	-48.000	11.455	-7.243;	1110	-48.000	11.767	-6.237;
1111	-48.000	11.848	-4.900;	1112	-48.000	11.939	-3.400;
1113	-48.000	12.146	0.000;	1114	-48.000	11.939	3.400;
1115	-48.000	11.848	4.900;	1116	-48.000	11.767	6.237;
1117	-48.000	11.455	7.243;	1118	-48.000	10.781	8.055;
1119	-48.000	9.849	8.543;	1120	-48.000	8.800	8.640;
1121	-48.000	7.600	8.514;	1122	-48.000	5.600	8.304;
1123	-48.000	4.926	8.233;	1124	-48.000	3.200	8.051;
1125	-48.000	0.000	7.715;				
1202	-45.600	3.200	-8.051;	1204	-45.600	5.600	-8.304;
1205	-45.600	7.781	-8.540;	1210	-45.600	11.949	-6.264;
1212	-45.600	12.122	-3.400;	1213	-45.600	12.947	0.000;
1214	-45.600	12.122	3.400;	1216	-45.600	11.949	6.264;
1221	-45.600	7.781	8.540;	1222	-45.600	5.600	8.304;
1223	-45.600	4.926	8.237;	1224	-45.600	3.200	8.051;
1225	-45.600	0.000	7.715;				
1301	-40.000	0.000	-7.715;	1302	-40.000	3.200	-8.051;
1303	-40.000	5.033	-8.244;	1304	-40.000	5.600	-8.304;
1305	-40.000	7.859	-8.541;	1306	-40.000	9.059	-8.667;
1307	-40.000	10.108	-8.571;	1308	-40.000	11.040	-8.080;
1309	-40.000	11.714	-7.270;	1310	-40.000	12.027	-6.264;
1311	-40.000	12.110	-4.900;	1312	-40.000	12.201	3.400;
1313	-40.000	12.407	0.000;	1314	-40.000	12.201	3.400;
1315	-40.000	12.110	4.900;	1316	-40.000	12.027	6.264;
1317	-40.000	11.714	7.270;	1318	-40.000	11.040	8.080;
1319	-40.000	10.108	8.571;	1320	-40.000	9.059	8.667;
1321	-40.000	7.859	8.541;	1322	-40.000	5.600	8.304;
1323	-40.000	5.033	8.244;	1324	-40.000	3.200	8.051;
1325	-40.000	0.000	7.715;				
1401	-32.000	0.000	-7.715;	1402	-32.000	3.200	-8.051;
1403	-32.000	5.151	-8.257;	1404	-32.000	5.600	-8.304;
1405	-32.000	8.071	-8.564;	1406	-32.000	9.271	-8.690;
1407	-32.000	10.320	-8.593;	1408	-32.000	11.253	-8.103;
1409	-32.000	11.926	-7.293;	1410	-32.000	12.239	-6.287;
1411	-32.000	12.323	-4.900;	1412	-32.000	12.414	-3.400;
1413	-32.000	12.621	0.000;	1414	-32.000	12.414	3.400;
1415	-32.000	12.323	4.900;	1416	-32.000	12.239	6.287;
1417	-32.000	11.926	7.293;	1418	-32.000	11.253	8.103;
1419	-32.000	10.320	8.593;	1420	-32.000	9.271	8.690;
1421	-32.000	8.071	8.564;	1422	-32.000	5.600	8.293;
1423	-32.000	5.151	8.257;	1424	-32.000	3.200	8.051;
1425	-32.000	0.000	7.715;				
1501	-24.000	0.000	-7.715;	1502	-24.000	3.200	-8.051;
1503	-24.000	5.242	-8.266;	1504	-24.000	5.500	-8.304;
1505	-24.000	8.236	-8.581;	1506	-24.000	9.436	-8.707;
1507	-24.000	10.485	-8.610;	1508	-24.000	11.418	-8.112;
1509	-24.000	12.091	-7.310;	1510	-24.000	12.404	-6.304;
1511	-24.000	12.489	-4.900;	1512	-24.000	12.580	-3.400;
1513	-24.000	12.787	0.000;	1514	-24.000	12.580	3.400;
1515	-24.000	12.489	4.900;	1516	-24.000	12.404	6.304;

1517	-24.000	12.091	7.310;	1518	-24.000	11.418	8.112;
1519	-24.000	10.485	8.610;	1520	-24.000	9.436	8.707;
1521	-24.000	8.236	8.581;	1522	-24.000	5.600	8.304;
1523	-24.000	5.242	8.266;	1524	-24.000	3.200	8.051;
1525	-24.000	0.000	7.715;				
1601	-16.000	0.000	-7.715;	1602	-16.000	3.200	-8.051;
1603	-16.000	5.308	-8.273;	1604	-16.000	5.600	-8.304;
1605	-16.000	8.354	-8.593;	1606	-16.000	9.554	-8.719;
1607	-16.000	10.603	-8.623;	1608	-16.000	11.536	-8.132;
1609	-16.000	12.209	-7.322;	1610	-16.000	12.522	-6.316;
1611	-16.000	12.608	-4.900;	1612	-16.000	12.699	-3.400;
1613	-16.000	12.905	0.000;	1614	-16.000	12.699	3.400;
1615	-16.000	12.608	4.900;	1616	-16.000	12.522	6.316;
1617	-16.000	12.209	7.322;	1618	-16.000	11.536	8.132;
1619	-16.000	10.603	8.623;	1620	-16.000	9.554	8.719;
1621	-16.000	8.354	8.593;	1622	-16.000	5.600	8.304;
1623	-16.000	5.308	8.273;	1624	-16.000	3.200	8.051;
1625	-16.000	0.000	7.715;				
1701	-8.000	0.000	-7.715;	1702	-8.000	3.200	-8.051;
1703	-8.000	5.347	-8.277;	1704	-8.000	5.600	-8.304;
1705	-8.000	8.425	-8.601;	1706	-8.000	9.625	-8.727;
1707	-8.000	10.674	-8.630;	1708	-8.000	11.606	-8.140;
1709	-8.000	12.280	-7.330;	1710	-8.000	12.592	-6.323;
1711	-8.000	12.678	-4.900;	1712	-8.000	12.770	-3.400;
1713	-8.000	12.976	0.000;	1714	-8.000	12.770	3.400;
1715	-8.000	12.678	4.900;	1716	-8.000	12.592	6.323;
1717	-8.000	12.280	7.330;	1718	-8.000	11.606	8.140;
1719	-8.000	10.674	8.630;	1720	-8.000	9.625	8.727;
1721	-8.000	8.425	8.601;	1722	-8.000	5.600	8.304;
1723	-8.000	5.347	8.277;	1724	-8.000	3.200	8.051;
1725	-8.000	0.000	7.715;				
1801	0.000	0.000	-7.715;	1802	0.000	3.200	-8.051;
1803	0.000	5.360	-8.278;	1804	0.000	5.600	-8.304;
1805	0.000	8.449	-8.603;	1806	0.000	9.649	-8.729;
1807	0.000	10.698	-8.633;	1808	0.000	11.630	-8.142;
1809	0.000	12.300	-7.332;	1810	0.000	12.616	-6.326;
1811	0.000	12.703	-4.900;	1812	0.000	12.794	-3.400;
1813	0.000	13.000	0.000;	1814	0.000	12.794	3.400;
1815	0.000	12.703	4.900;	1816	0.000	12.616	6.326;
1817	0.000	12.300	7.332;	1818	0.000	11.630	8.142;
1819	0.000	10.698	8.633;	1820	0.000	9.649	8.729;
1821	0.000	8.449	8.603;	1822	0.000	5.600	8.304;
1823	0.000	5.360	8.278;	1824	0.000	3.200	8.051;
1825	0.000	0.000	7.715;				
1901	8.000	0.000	-7.715;	1902	8.000	3.200	-8.051;
1903	8.000	5.347	-8.277;	1904	8.000	5.600	-8.304;
1905	8.000	8.425	-8.601;	1906	8.000	9.625	-8.727;
1907	8.000	10.674	-8.630;	1908	8.000	11.606	-8.140;
1909	8.000	12.280	-7.330;	1910	8.000	12.592	-6.323;
1911	8.000	12.678	-4.900;	1912	8.000	12.770	-3.400;
1913	8.000	12.976	0.000;	1914	8.000	12.770	3.400;
1915	8.000	12.678	4.900;	1916	8.000	12.592	6.323;
1917	8.000	12.280	7.330;	1918	8.000	11.606	8.140;
1919	8.000	10.674	8.630;	1920	8.000	9.625	8.727;
1921	8.000	8.425	8.601;	1922	8.000	5.600	8.304;
1923	8.000	5.347	8.277;	1924	8.000	3.200	8.051;
1925	8.000	0.000	7.715;				
2001	16.000	0.000	-7.715;	2002	16.000	3.200	-8.051;
2003	16.000	5.308	-8.273;	2004	16.000	5.600	-8.304;
2005	16.000	8.354	-8.593;	2006	16.000	9.554	-8.719;
2007	16.000	10.603	-8.623;	2008	16.000	11.536	-8.132;
2009	16.000	12.209	-7.322;	2010	16.000	12.522	-6.316;
2011	16.000	12.608	-4.900;	2012	16.000	12.699	-3.400;
2013	16.000	12.905	0.000;	2014	16.000	12.699	3.400;
2015	16.000	12.608	4.900;	2016	16.000	12.522	6.316;
2017	16.000	12.209	7.322;	2018	16.000	11.536	8.132;
2019	16.000	10.603	8.623;	2020	16.000	9.554	8.719;
2021	16.000	8.354	8.593;	2022	16.000	5.600	8.304;

4401 1016 1116; 4402 1116 1216; 4403 1216 1316; 4404 1316 1416;
4405 1416 1516; 4406 1516 1616; 4407 1616 1716; 4408 1716 1816;
4409 1816 1916; 4410 1916 2016; 4411 2016 2116; 4412 2116 2216;
4413 2216 2316; 4414 2316 2416; 4415 2416 2516; 4416 2516 2616;

*SIDE BEAM
5001 1002 1102; 5002 1102 1202; 5003 1202 1302; 5004 1302 1402;
5005 1402 1502; 5006 1502 1602; 5007 1602 1702; 5008 1702 1802;
5009 1802 1902; 5010 1902 2002; 5011 2002 2102; 5012 2102 2202;
5013 2202 2302; 5014 2302 2402; 5015 2402 2502; 5016 2502 2602;

5101 1004 1104; 5102 1104 1204; 5103 1204 1304; 5104 1304 1404;
5105 1404 1504; 5106 1504 1604; 5107 1604 1704; 5108 1704 1804;
5109 1804 1904; 5110 1904 2004; 5111 2004 2104; 5112 2104 2204;
5113 2204 2304; 5114 2304 2404; 5115 2404 2504; 5116 2504 2604;

5201 1005 1105; 5202 1105 1205; 5203 1205 1305; 5204 1305 1405;
5205 1405 1505; 5206 1505 1605; 5207 1605 1705; 5208 1705 1805;
5209 1805 1905; 5210 1905 2005; 5211 2005 2105; 5212 2105 2205;
5213 2205 2305; 5214 2305 2405; 5215 2405 2505; 5216 2505 2605;

5301 1021 1121; 5302 1121 1221; 5303 1221 1321; 5304 1321 1421;
5305 1421 1521; 5306 1521 1621; 5307 1621 1721; 5308 1721 1821;
5309 1821 1921; 5310 1921 2021; 5311 2021 2121; 5312 2121 2221;
5313 2221 2321; 5314 2321 2421; 5315 2421 2521; 5316 2521 2621;

5401 1022 1122; 5402 1122 1222; 5403 1222 1322; 5404 1322 1422;
5405 1422 1522; 5406 1522 1622; 5407 1622 1722; 5408 1722 1822;
5409 1822 1922; 5410 1922 2022; 5411 2022 2122; 5412 2122 2222;
5413 2222 2322; 5414 2322 2422; 5415 2422 2522; 5416 2522 2622;

5501 1024 1124; 5502 1124 1224; 5503 1224 1324; 5504 1324 1424;
5505 1424 1524; 5506 1524 1624; 5507 1624 1724; 5508 1724 1824;
5509 1824 1924; 5510 1924 2024; 5511 2024 2124; 5512 2124 2224;
5513 2224 2324; 5514 2324 2424; 5515 2424 2524; 5516 2524 2624;

*OFFICE

1230 1223 1231; 1231 1231 1232 1233; 1330 1323 1331;
1230 1331 1332 1333; 1430 1423 1431; 1431 1431 1432 1433;
1530 1523 1531; 1531 1531 1532 1533; 1630 1623 1631;
1631 1631 1632 1633; 1730 1723 1731; 1731 1731 1732 1733;
1830 1823 1831; 1831 1831 1832 1833; 1930 1923 1931;
1931 1931 1932 1933; 2030 2023 2031; 2031 2031 2032 2033;
2130 2123 2131; 2131 2131 2132 2133; 2230 2223 2231;
2231 2231 2232 2233; 2330 2323 2331; 2331 2331 2332 2333;
2430 2423 2431; 2431 2431 2432 2433;

*SECONDARY BEAM

6001 1231 1331; 6002 1331 1431; 6003 1431 1531; 6004 1531 1631;
6005 1631 1731; 6006 1731 1831; 6007 1831 1931; 6008 1931 2031;
6009 2031 2131; 6010 2131 2231; 6011 2231 2331; 6012 2331 2431;

6101 1232 1332; 6102 1332 1432; 6103 1432 1532; 6104 1532 1632;
6105 1632 1732; 6106 1732 1832; 6107 1832 1932; 6108 1932 2032;
6109 2032 2132; 6110 2132 2232; 6111 2232 2332; 6112 2332 2432;

6201 1233 1333; 6202 1333 1433; 6203 1433 1533; 6204 1533 1633;
6205 1633 1733; 6206 1733 1833; 6207 1833 1933; 6208 1933 2033;
6209 2033 2133; 6210 2133 2233; 6211 2233 2333; 6212 2333 2433;

*PERIMETER COLUMN

110 3011 3021; 111 3021 3031; 112 3031 3041; 113 3041 1111;
120 3013 3023; 121 3023 3033; 122 3033 3043; 123 3043 1113;
130 3015 3025; 131 3025 3035; 132 3035 3045; 133 3045 1115;
210 3111 3121; 211 3121 3131; 212 3131 3141; 213 3141 2511;
220 3113 3123; 221 3123 3133; 222 3133 3143; 223 3143 2513;
230 3115 3125; 231 3125 3135; 232 3135 3145; 233 3145 2515;

*MAIN HALL - ROOF BRACING

7001 1005 1101; 7002 1105 1010; 7003 1012 1113; 7004 1112 1013;
7005 1013 1114; 7006 1113 1014; 7007 1016 1121; 7008 1116 1021;

7101 1105 1310; 7102 1305 1110; 7103 1110 1312; 7104 1310 1112;
7105 1112 1313; 7106 1312 1113; 7107 1113 1314; 7108 1313 1114;
7109 1114 1316; 7110 1314 1116; 7111 1116 1321; 7112 1316 1121;

7201 1805 1910; 7202 1905 1810; 7203 1810 1912; 7204 1910 1812;
7205 1812 1913; 7206 1912 1813; 7207 1813 1914; 7208 1913 1814;
7209 1814 1916; 7210 1914 1816; 7211 1816 1921; 7212 1916 1821;

7301 1905 2010; 7302 2005 1910; 7303 1910 2012; 7304 2010 1912;
7305 1912 2013; 7306 2012 1913; 7307 1913 2014; 7308 2013 1914;
7309 1914 2016; 7310 2014 1916; 7311 1916 2021; 7312 2016 1921;

7401 2305 2510; 7402 2505 2310; 7403 2310 2512; 7404 2510 2312;
7405 2312 2513; 7406 2512 2313; 7407 2313 2514; 7408 2513 2314;
7409 2314 2516; 7410 2514 2316; 7411 2316 2521; 7412 2516 2321;

7501 2505 2610; 7502 2605 2510; 7503 2512 2613; 7504 2612 2513;
7505 2513 2614; 7506 2613 2514; 7507 2516 2621; 7508 2616 2521;

*VERTICAL BRACING
7601 1101 1302; 7602 1301 1102; 7603 1102 1304; 7604 1302 1104;
7605 1104 1305; 7606 1304 1105; 7607 1121 1322; 7608 1321 1122;
7609 1122 1224; 7610 1222 1124; 7611 1124 1225; 7612 1224 1125;
7613 1222 1324; 7614 1322 1224; 7615 1224 1325; 7616 1324 1225;

7701 1801 1902; 7702 1901 1802; 7703 1802 1904; 7704 1902 1804;
7705 1804 1905; 7706 1904 1805; 7707 1821 1922; 7708 1921 1822;
7709 1822 1924; 7710 1922 1824; 7711 1824 1925; 7712 1924 1825;

7801 1901 2002; 7802 2001 1902; 7803 1902 2004; 7804 2002 1904;
7805 1904 2005; 7806 2004 1905; 7807 1921 2022; 7808 2021 1922;
7809 1922 2024; 7810 2022 1924; 7811 1924 2025; 7812 2024 1925;

7901 2301 2502; 7902 2501 2302; 7903 2302 2504; 7904 2502 2304;
7905 2304 2505; 7906 2504 2305; 7907 2321 2522; 7908 2521 2322;
7909 2322 2424; 7910 2422 2324; 7911 2324 2425; 7912 2424 2325;
7913 2422 2524; 7914 2522 2424; 7915 2424 2525; 7916 2524 2425;

*PERIMETER BRACING
8001 1101 3021; 8002 3011 1102; 8003 1102 3031; 8004 3021 1105;
8005 1105 3041; 8006 3031 1106;
8007 1124 3035; 8008 3025 1121; 8009 1121 3045; 8010 3035 1120;
8011 1102 3021; 8012 1105 3031; 8013 1106 3041;
8014 1120 3045; 8015 1121 3035; 8016 1124 3025;

8101 2501 3121; 8102 3111 2502; 8103 2502 3131; 8104 3121 2505;
8105 2505 3141; 8106 3131 2506;
8107 2524 3135; 8108 3125 2521; 8109 2521 3145; 8110 3135 2520;
8111 2502 3121; 8112 2505 3131; 8113 2506 3141;
8114 2520 3145; 8115 2521 3135; 8116 2524 3125;

*OFFICE BRACING
9001 1231 1332; 9002 1331 1232; 9003 1232 1333; 9004 1332 1233;
9005 1233 1334; 9006 1333 1234;

9101 1831 1932; 9102 1931 1832; 9103 1832 1933; 9104 1932 1833;
9105 1833 1934; 9106 1933 1834;

9201 1931 2032; 9202 2031 1932; 9203 1932 2033; 9204 2032 1933;

9301 2331 2432; 9302 2431 2332; 9303 2332 2433; 9304 2432 2333;
9305 2333 2434; 9306 2433 2334;

9401 1223 1234; 9402 1225 1233; 9403 2423 2434; 9404 2425 2433;

DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03

END DEFINE MATERIAL

MEMBER PROPERTY BRITISH

*MAIN HALL - MAIN FRAME

1101 1301 1401 1501 1601 1701 1801 1901 2001 2101 2201 2301 2501 -
TAPERED 0.7 0.010 0.8 0.25 0.015 0.25 0.015
1102 1302 1402 1502 1602 1702 1802 1902 2002 2102 2202 2302 2502 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1103 1303 1403 1503 1603 1703 1803 1903 2003 2103 2203 2303 2503 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1104 1304 1404 1504 1604 1704 1804 1904 2004 2104 2204 2304 2504 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1105 1305 1405 1505 1605 1705 1805 1905 2005 2105 2205 2305 2505 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1106 1306 1406 1506 1606 1706 1806 1906 2006 2106 2206 2306 2506 -
TAPERED 0.8 0.010 0.75 0.25 0.015 0.25 0.015
1107 1307 1407 1507 1607 1707 1807 1907 2007 2107 2207 2307 2507 -
TAPERED 0.75 0.010 0.7 0.25 0.015 0.25 0.015
1108 1308 1408 1508 1608 1708 1808 1908 2008 2108 2208 2308 2508 -
TAPERED 0.7 0.010 0.65 0.25 0.015 0.25 0.015
1109 1309 1409 1509 1609 1709 1809 1909 2009 2109 2209 2309 2509 -
TAPERED 0.65 0.010 0.6 0.25 0.015 0.25 0.015
1110 TO 1115 -
1510 TO 1515 -
1610 TO 1615 1710 TO 1715 1810 TO 1815 1910 TO 1915 -
2010 TO 2015 2110 TO 2115 -
2510 TO 2515 TAPERED 0.6 0.008 0.6 0.2 0.012 0.2 0.012
*

1310 1410 2210 2310 -
TAPERED 0.6 0.008 0.7 0.2 0.012 0.2 0.012

*

1311 1411 2211 2311 -
TAPERED 0.7 0.008 0.8 0.2 0.012 0.2 0.012

*

1312 1412 2212 2312 -
TAPERED 0.8 0.008 1.0 0.2 0.012 0.2 0.012

*

1313 1413 2213 2313 -
TAPERED 1.0 0.008 0.8 0.2 0.012 0.2 0.012

*

1314 1414 2214 2314 -
TAPERED 0.8 0.008 0.7 0.2 0.012 0.2 0.012

*

1315 1415 2215 2315 -
TAPERED 0.7 0.008 0.6 0.2 0.012 0.2 0.012

*

1116 1316 1416 1516 1616 1716 1816 1916 2016 2116 2216 2316 2516 -
TAPERED 0.6 0.010 0.65 0.25 0.015 0.25 0.015
1117 1317 1417 1517 1617 1717 1817 1917 2017 2117 2217 2317 2517 -
TAPERED 0.65 0.010 0.7 0.25 0.015 0.25 0.015
1118 1318 1418 1518 1618 1718 1818 1918 2018 2118 2218 2318 2518 -
TAPERED 0.7 0.010 0.75 0.25 0.015 0.25 0.015
1119 1319 1419 1519 1619 1719 1819 1919 2019 2119 2219 2319 2519 -
TAPERED 0.75 0.010 0.8 0.25 0.015 0.25 0.015
1120 1320 1420 1520 1620 1720 1820 1920 2020 2120 2220 2320 2520 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1121 1321 1421 1521 1621 1721 1821 1921 2021 2121 2221 2321 2521 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1122 1322 1422 1522 1622 1722 1822 1922 2022 2122 2222 2322 2522 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1123 1323 1423 1523 1623 1723 1823 1923 2023 2123 2223 2323 2523 -

TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1124 1324 1424 1524 1624 1724 1824 1924 2024 2124 2224 2324 2524 -
TAPERED 0.8 0.010 0.7 0.25 0.015 0.25 0.015
1222 TO 1224 -
2422 TO 2424 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
*MAIN HALL SECONDARY BEAMS
4001 4016 4101 4116 4201 4216 4301 4316 4401 4416 -
TAPERED 0.3 0.006 0.3 0.15 0.01 0.15 0.01
4002 TO 4015 4102 TO 4115 4202 TO 4215 4302 TO 4315 -
4402 TO 4415 TAPERED 0.4 0.006 0.4 0.15 0.01 0.15 0.01
*SIDE BEAM
5001 TO 5016 5101 TO 5116 5201 TO 5216 5301 TO 5316 5401 TO 5416 -
5501 TO 5516 TAPERED 0.3 0.010 0.3 0.25 0.15 0.25 0.15
*OFFICE MAIN ROOF BEAM
1230 TO 1232 1330 TO 1332 1430 TO 1432 1530 TO 1532 -
1630 TO 1632 1730 TO 1732 1830 TO 1832 1930 TO 1932 -
2030 TO 2032 2130 TO 2132 2230 TO 2232 2330 TO 2332 -
2430 TO 2432 TAPERED 0.4 0.006 0.4 0.15 0.01 0.15 0.01
1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
2433 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
*OFFICE SECONDARY BEAM
6001 TO 6012 6101 TO 6112 -
6201 TO 6212 TAPERED 0.4 0.006 0.4 0.15 0.01 0.15 0.01
*PERIMETER COLUMN
110 TO 113 130 TO 133 210 TO 213 230 TO 233 -
TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
120 TO 123 220 TO 223 TAPERED 0.3 0.010 0.3 0.3 0.015 0.3 0.015
*BRACING
7001 TO 7008 7101 TO 7112 7201 TO 7212 7301 TO 7312 -
7401 TO 7412 7501 TO 7508 7601 TO 7616 7701 TO 7712 -
7801 TO 7812 7901 TO 7916 8001 TO 8016 8101 TO 8116 -
9001 TO 9006 9101 TO 9106 9201 TO 9204 9301 TO 9306 TABLE ST UA90X90X6
9401 TO 9404 TABLE ST UA90X90X6
MEMBER RELEASES
4004 TO 4013 4104 TO 4113 4204 TO 4213 4304 TO 4313 -
4404 TO 4413 BOTH MX MY MZ
4003 4103 4203 4303 4403 END MX MY MZ
4014 4114 4214 4314 4414 START MX MY MZ
6001 TO 6012 6101 TO 6112 6201 TO 6212 BOTH MX MY MZ
1230 1330 1430 1530 1630 1730 1830 1930 2030 2130 -
2230 2330 2430 START MX MY MZ
MEMBER CABLE
7001 TO 7008 7101 TO 7112 7201 TO 7212 7301 TO 7312 -
7401 TO 7412 7501 TO 7508 7601 TO 7616 7701 TO 7712 -
7801 TO 7812 7901 TO 7916 8001 TO 8016 8101 TO 8116 -
9001 TO 9006 9101 TO 9106 9201 TO 9204 9301 TO 9306 -
9401 TO 9404 TENSION 5
CONSTANTS
MATERIAL STEEL ALL
BETA 90 MEMB 5001 TO 5016 5101 TO 5116 5201 TO 5216 5301 TO 5316 -
5401 TO 5416 5501 TO 5516
SUPPORTS
1101 1125 1225 1301 1325 1401 1425 1501 1525 1601 1625 -
1701 1725 1801 1825 1901 1925 2001 2025 2101 2125 2201 2225 -
2301 2325 2425 2501 2525 PINNED
1234 1334 1434 1534 1634 1734 1834 1934 -
2034 2134 2234 2334 2434 PINNED
3011 3013 3015 3111 3113 3115 PINNED
LOAD 1 SELFWEIGHT
SELFWEIGHT Y -1
LOAD 2 SDL

MEMBER LOAD
*MAIN HALL
4001 TO 4016 4401 TO 4416 UNI GY -4.62
4101 TO 4116 4301 TO 4316 UNI GY -3.16
4201 TO 4216 UNI GY -3.41
5001 TO 5016 5501 TO 5516 UNI GY -2.19
5101 TO 5116 5401 TO 5416 UNI GY -1.27
5201 TO 5216 5301 TO 5316 UNI GY -1.82
*OFFICE
6001 TO 6012 UNI GY -1.53
6101 TO 6112 UNI GY -2.38
6201 TO 6212 UNI GY -3.88

*AHU PLATFORM LOAD
1305 1405 2205 2305 CON GY -20.3 0.8
1312 1412 2212 2312 CON GY -20.3 0.275
1313 1413 2213 2313 CON GY -20.3 3.125
1320 1420 2220 2320 CON GY -20.3 0.4

LOAD 3 LL
MEMBER LOAD
*MAIN HALL
4001 TO 4016 4401 TO 4416 UNI GY -2.77
4101 TO 4116 4301 TO 4316 UNI GY -1.90
4201 TO 4216 UNI GY -2.05
*OFFICE
6001 TO 6012 UNI GY -0.92
6101 TO 6112 UNI GY -1.43
6201 TO 6212 UNI GY -0.71
*AIRCON LOAD
4204 TO 4213 CON GY -5
4203 CON GY -5 1.6
4214 CON GY -5 4.0
6101 TO 6112 CON GY -5

*AHU PLATFORM LOAD
1305 1405 2205 2305 CON GY -26.3 0.8
1312 1412 2212 2312 CON GY -26.3 0.275
1313 1413 2213 2313 CON GY -26.3 3.125
1320 1420 2220 2320 CON GY -26.3 0.4

*CRANE LOAD
JOINT LOAD
1804 FY -236.8
1804 FZ 35.5

LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
MEMBER LOAD
*MAIN HALL
4001 TO 4016 UNI GY 3.88
4101 TO 4116 UNI GY 2.65
4201 TO 4216 UNI GY 1.84
4401 TO 4416 UNI GY 0.76
4301 TO 4316 UNI GY 1.11
5001 TO 5016 UNI GZ 6.18
5101 TO 5116 UNI GZ 3.57
5201 TO 5216 UNI GZ 5.13
5301 TO 5316 UNI GZ 0.91
5401 TO 5416 UNI GZ 0.63
5501 TO 5516 UNI GZ 1.10
*OFFICE
6001 TO 6012 UNI GY 1.32
6101 TO 6112 UNI GY 1.62
6201 TO 6212 UNI GY 0.59
1233 2433 UNI GZ 1.00
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 2.00

LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
MEMBER LOAD
4001 TO 4016 UNI GY 6.75
4101 TO 4116 UNI GY 4.61
4201 TO 4216 UNI GY 3.34

4401 TO 4416 UNI GY 1.58
4301 TO 4316 UNI GY 2.31
5001 TO 5016 UNI GZ 3.46
5101 TO 5116 UNI GZ 2.00
5201 TO 5216 UNI GZ 2.88
5301 TO 5316 UNI GZ 3.17
5401 TO 5416 UNI GZ 2.20
5501 TO 5516 UNI GZ 3.81
*OFFICE
6001 TO 6012 UNI GY 2.26
6101 TO 6112 UNI GY 2.36
6201 TO 6212 UNI GY 0.59
1233 2433 UNI GZ 3.48
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 6.96

LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
MEMBER LOAD
4001 TO 4016 UNI GY -2.86
4101 TO 4116 UNI GY -1.96
4201 TO 4216 UNI GY -2.11
4401 TO 4416 UNI GY -1.96
4301 TO 4316 UNI GY -2.86
5001 TO 5016 UNI GZ -2.72
5101 TO 5116 UNI GZ -1.57
5201 TO 5216 UNI GZ -2.26
5301 TO 5316 UNI GZ 2.26
5401 TO 5416 UNI GZ 1.57
5501 TO 5516 UNI GZ 2.72
*OFFICE
6001 TO 6012 UNI GY -0.95
6101 TO 6112 UNI GY -1.48
6201 TO 6212 UNI GY -0.73
1233 2433 UNI GZ 2.48
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 4.96

LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
MEMBER LOAD
4001 TO 4016 UNI GY 5.17
4101 TO 4116 UNI GY 3.54
4201 TO 4216 UNI GY 3.82
4401 TO 4416 UNI GY 3.54
4301 TO 4316 UNI GY 5.17
5001 TO 5016 UNI GZ -5.43
5101 TO 5116 UNI GZ -3.14
5201 TO 5216 UNI GZ -4.51
5301 TO 5316 UNI GZ 4.51
5401 TO 5416 UNI GZ 3.14
5501 TO 5516 UNI GZ 5.43
*OFFICE
6001 TO 6012 UNI GY 1.71
6101 TO 6112 UNI GY 2.67
6201 TO 6212 UNI GY 1.32
1233 2433 UNI GZ 4.96
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 9.92

LOAD 21 EQ1 (EARTHQUAKE LOAD)
JOINT LOAD
1109 1115 1309 1315 1409 1415 1509 1515 1609 1615 1709 1715 -
1809 1815 1909 1915 2009 2015 2109 2115 2209 2215 2309 2315 -
2509 2515 FX 28.5
1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
2433 FX 19.1

LOAD 22 EQ1 (EARTHQUAKE LOAD)
JOINT LOAD
1109 1115 1309 1315 1409 1415 1509 1515 1609 1615 1709 1715 -
1809 1815 1909 1915 2009 2015 2109 2115 2209 2215 2309 2315 -
2509 2515 FZ 28.5
1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
2433 FZ 19.1

LOAD COMB 1001 1.4DL + 1.6LL
1 1.4 2 1.4 3 1.6

LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
1 1.2 2 1.2 3 1.2 11 1.2
LOAD COMB 1111 1.4DL + 1.4WL1
1 1.4 2 1.4 11 1.4
LOAD COMB 1211 1.0DL + 1.4WL1
1 1.0 2 1.0 11 1.4
LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
1 1.2 2 1.2 3 1.2 12 1.2
LOAD COMB 1112 1.4DL + 1.4WL2
1 1.4 2 1.4 12 1.4
LOAD COMB 1212 1.0DL + 1.4WL2
1 1.0 2 1.0 12 1.4
LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
1 1.2 2 1.2 3 1.2 13 1.2
LOAD COMB 1113 1.4DL + 1.4WL3
1 1.4 2 1.4 13 1.4
LOAD COMB 1213 1.0DL + 1.4WL3
1 1.0 2 1.0 13 1.4
LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
1 1.2 2 1.2 3 1.2 14 1.2
LOAD COMB 1114 1.4DL + 1.4WL4
1 1.4 2 1.4 14 1.4
LOAD COMB 1214 1.0DL + 1.4WL4
1 1.0 2 1.0 14 1.4

LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
1 1.2 2 1.2 3 0.5 21 1.0
LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
1 1.2 2 1.2 3 0.5 22 1.0
LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
1 1.2 2 1.2 3 0.5 21 -1.0
LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
1 1.2 2 1.2 3 0.5 22 -1.0
LOAD COMB 1221 0.9DL + 1.0EQ1
1 0.9 2 0.9 21 1.0
LOAD COMB 1222 0.9DL + 1.0EQ2
1 0.9 2 0.9 22 1.0
LOAD COMB 1321 0.9DL - 1.0EQ1
1 0.9 2 0.9 21 -1.0
LOAD COMB 1322 0.9DL - 1.0EQ2
1 0.9 2 0.9 22 -1.0
*SLS
LOAD COMB 2001 1.0DL + 1.0LL
1 1.0 2 1.0 3 1.0
LOAD COMB 2002 1.0DL + 0.5LL
1 1.0 2 1.0 3 0.5
LOAD COMB 2011 DL + LL + WL1
1 1.0 2 1.0 3 1.0 11 1.0
LOAD COMB 2012 DL + LL + WL2
1 1.0 2 1.0 3 1.0 12 1.0
LOAD COMB 2013 DL + LL + WL3
1 1.0 2 1.0 3 1.0 13 1.0
LOAD COMB 2014 DL + LL + WL4
1 1.0 2 1.0 3 1.0 14 1.0

PERFORM ANALYSIS
LOAD LIST 1001 1011 TO 1014 1021 1022 1111 TO 1114 1121 1122 -
1211 TO 1214 1221 1222 1321 1322
PRINT SUPPORT REACTION ALL
PRINT FORCE ENVELOPE NSECTION 5 ALL
PARAMETER 1
CODE BS950
SBLT 1.0
PY 345000 ALL
CHECK CODE ALL
PARAMETER 2
CODE BS950
TRACK 2 ALL
SBLT 1.0

PY 345000 ALL
CHECK CODE MEM 1401 TO 1424
LOAD LIST 2001 2002 2011 TO 2014
PRINT JOINT DISPLACEMENTS ALL
FINISH

STAAD Output File - Main Frame Steel Members Check

795. PARAMETER 1
796. CODE BS950
797. SBLT 1.0
798. PY 345000 ALL
799. CHECK CODE ALL
•STEEL DESIGN •

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1301	TAPERED	PASS	BS-4.7 (C)	0.726	1021
1302	TAPERED	498.39 C	77.31	160.07	3.22
1303	TAPERED	PASS	BS-4.7 (C)	0.412	1021
1304	TAPERED	410.52 C	36.39	-145.02	0.00
1305	TAPERED	PASS	BS-4.7 (C)	0.483	1021
1306	TAPERED	407.13 C	36.34	262.38	0.57
1307	TAPERED	PASS	BS-4.7 (C)	0.708	1021
1308	TAPERED	333.65 C	62.29	372.18	2.27
1309	TAPERED	PASS	BS-4.7 (C)	0.499	1021
1310	TAPERED	225.14 C	27.43	-376.36	0.00
1311	TAPERED	PASS	BS-4.7 (C)	0.509	1013
1312	TAPERED	260.40 C	-0.06	-553.80	1.05
1313	TAPERED	PASS	BS-4.7 (C)	0.504	1001
1314	TAPERED	244.19 C	0.09	551.22	0.00
1315	TAPERED	PASS	BS-4.7 (C)	0.479	1122
1316	TAPERED	146.56 C	0.11	498.14	0.00
1317	TAPERED	PASS	BS-4.7 (C)	0.379	1122
1318	TAPERED	61.31 C	-0.02	370.56	0.00
1319	TAPERED	PASS	BS-4.7 (C)	0.378	1022
1320	TAPERED	29.17 C	0.35	278.02	1.37
1321	TAPERED	PASS	BS-4.7 (C)	0.548	1013
1322	TAPERED	16.74 C	0.98	484.20	1.50
1323	TAPERED	PASS	BS-4.3.6	0.540	1011
1324	TAPERED	58.56 C	-0.80	369.88	0.28
1325	TAPERED	PASS	BS-4.3.6	0.545	1013
1326	TAPERED	10.16 C	0.46	616.21	0.00
1327	TAPERED	PASS	BS-4.7 (C)	0.521	1021
1328	TAPERED	7.88 C	-28.20	146.46	1.50
1329	TAPERED	PASS	BS-4.7 (C)	0.534	1021
1330	TAPERED	7.90 C	28.20	-146.46	0.00
1331	TAPERED	PASS	BS-4.7 (C)	0.447	1011
1332	TAPERED	100.47 C	0.02	-430.56	1.05
1333	TAPERED	PASS	BS-4.7 (C)	0.555	1011
1334	TAPERED	167.29 C	-0.11	-578.00	1.05
1335	TAPERED	PASS	BS-4.7 (C)	0.577	1011
1336	TAPERED	209.49 C	-0.24	-647.76	1.05
1337	TAPERED	PASS	BS-4.7 (C)	0.579	1011
1338	TAPERED	218.99 C	0.21	647.76	0.00
1339	TAPERED	PASS	BS-4.7 (C)	0.534	1012
1340	TAPERED	192.15 C	-0.34	-652.93	0.00
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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
2201	TAPERED	PASS	BS-4.7 (C)	0.719	1121
2202	TAPERED	430.09 C	-76.46	179.81	3.22
2203	TAPERED	PASS	BS-4.7 (C)	0.443	1013
2204	TAPERED	476.76 C	-0.29	487.17	1.96

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1321	TAPERED	PASS	BS-4.7 (C)	0.646	1021
1322	TAPERED	334.74 C	-52.50	-369.67	0.00
1323	TAPERED	PASS	BS-4.7 (C)	0.448	1013
1324	TAPERED	452.13 C	4.81	-478.72	0.00
1325	TAPERED	PASS	BS-4.7 (C)	0.426	1013
1326	TAPERED	496.12 C	1.81	-442.11	0.00
1327	TAPERED	PASS	BS-4.7 (C)	0.436	1021
1328	TAPERED	521.40 C	-32.64	-141.59	0.00
1329	TAPERED	PASS	BS-4.7 (C)	0.720	1021
1330	TAPERED	429.57 C	76.48	179.99	3.22
1331	TAPERED	PASS	BS-4.7 (C)	0.443	1013
1332	TAPERED	476.95 C	0.18	487.56	1.96
1333	TAPERED	PASS	BS-4.7 (C)	0.495	1021
1334	TAPERED	351.37 C	38.30	284.52	0.45
1335	TAPERED	PASS	BS-4.7 (C)	0.599	1021
1336	TAPERED	279.51 C	42.07	364.38	2.48
1337	TAPERED	PASS	BS-4.7 (C)	0.554	1013
1338	TAPERED	280.08 C	-0.03	657.93	1.21
1339	TAPERED	PASS	BS-4.7 (C)	0.598	1122
1340	TAPERED	214.53 C	-0.09	-673.30	1.05
1341	TAPERED	PASS	BS-4.7 (C)	0.596	1122
1342	TAPERED	203.33 C	0.11	673.30	0.00
1343	TAPERED	PASS	BS-4.7 (C)	0.574	1122
1344	TAPERED	160.12 C	0.06	600.23	0.00
1345	TAPERED	PASS	BS-4.7 (C)	0.461	1122
1346	TAPERED	66.45 C	0.00	451.99	0.00
1347	TAPERED	PASS	BS-4.8.2.2	0.541	1112
1348	TAPERED	11.53 T	0.08	-334.62	0.00
1349	TAPERED	PASS	BS-4.8.2.2	0.626	1013
1350	TAPERED	9.89 T	0.75	559.87	1.50
1351	TAPERED	PASS	BS-4.3.6	0.616	1011
1352	TAPERED	41.46 C	-0.66	439.41	0.28
1353	TAPERED	PASS	BS-4.3.6	0.629	1013
1354	TAPERED	17.42 T	0.35	714.71	0.00
1355	TAPERED	PASS	BS-4.3.6	0.560	1011
1356	TAPERED	49.10 C	-0.57	157.95	0.00
1357	TAPERED	PASS	BS-4.7 (C)	0.576	1211
1358	TAPERED	39.90 C	0.12	-349.53	1.39
1359	TAPERED	PASS	BS-4.7 (C)	0.552	1011
1360	TAPERED	107.36 C	0.01	-534.96	1.05
1361	TAPERED	PASS	BS-4.7 (C)	0.667	1011
1362	TAPERED	181.36 C	-0.08	-699.57	1.05
1363	TAPERED	PASS	BS-4.7 (C)	0.688	1011
1364	TAPERED	228.90 C	-0.16	-779.17	1.05
1365	TAPERED	PASS	BS-4.7 (C)	0.691	1011
1366	TAPERED	240.54 C	0.14	779.17	0.00
1367	TAPERED	PASS	BS-4.7 (C)	0.640	1012
1368	TAPERED	211.55 C	-0.23	-789.35	0.00
1369	TAPERED	PASS	BS-4.3.6	0.600	1112
1370	TAPERED	255.58 C	-0.49	686.57	0.00
1371	TAPERED	PASS	BS-4.7 (C)	0.514	1013
1372	TAPERED	485.43 C	-0.55	-599.41	0.00
1373	TAPERED	PASS	BS-4.7 (C)	0.522	1013
1374	TAPERED	532.47 C	-0.75	-572.73	0.00
1375	TAPERED	PASS	BS-4.7 (C)	0.454	1021
1376	TAPERED	463.13 C	-33.98	-179.98	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
2201	TAPERED	PASS	BS-4.7 (C)	0.719	1121
2202	TAPERED	430.09 C	-76.46	179.81	3.22
2203	TAPERED	PASS	BS-4.7 (C)	0.443	1013
2204	TAPERED	476.76 C	-0.29	487.17	1.96

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2203	TAPERED	PASS	BS-4.7	(C)	0.497	1121
2204	TAPERED	351.62 C	BS-4.7	(C)	284.40	0.45
2205	TAPERED	PASS	BS-4.7	(C)	0.602	1121
2205	TAPERED	279.71 C	BS-4.7	(C)	364.70	2.48
2206	TAPERED	PASS	BS-4.7	(C)	0.554	1013
2206	TAPERED	280.01 C	BS-4.7	(C)	657.06	1.21
2207	TAPERED	PASS	BS-4.7	(C)	0.599	1122
2207	TAPERED	214.61 C	BS-4.7	(C)	-673.78	1.05
2208	TAPERED	PASS	BS-4.7	(C)	0.596	1122
2208	TAPERED	203.44 C	BS-4.7	(C)	673.78	0.00
2209	TAPERED	PASS	BS-4.7	(C)	0.574	1122
2209	TAPERED	160.24 C	BS-4.7	(C)	600.76	0.00
2210	TAPERED	PASS	BS-4.7	(C)	0.461	1122
2210	TAPERED	66.56 C	BS-4.7	(C)	452.52	0.00
2211	TAPERED	PASS	BS-4.8.2.2	(C)	0.544	1112
2211	TAPERED	11.40 T	BS-4.8.2.2	(C)	-335.93	0.00
2211	TAPERED	PASS	BS-4.8.2.2	(C)	0.625	1013
2211	TAPERED	9.75 T	BS-4.8.2.2	(C)	-0.66	1.50
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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
2212	TAPERED	PASS	BS-4.3.6	0.616	1011
2213	TAPERED	41.60 C	0.57	440.04	0.28
2214	TAPERED	PASS	BS-4.3.6	0.629	1013
2215	TAPERED	17.27 T	-0.33	714.62	0.00
2216	TAPERED	PASS	BS-4.3.6	0.560	1011
2217	TAPERED	49.28 C	0.57	157.02	0.00
2218	TAPERED	PASS	BS-4.7 (C)	0.578	1211
2219	TAPERED	40.04 C	-0.14	-351.01	1.39
2220	TAPERED	PASS	BS-4.7 (C)	0.554	1011
2221	TAPERED	107.58 C	-0.02	-536.73	1.05
2222	TAPERED	PASS	BS-4.7 (C)	0.669	1011
2223	TAPERED	181.63 C	0.08	-701.42	1.05
2224	TAPERED	PASS	BS-4.7 (C)	0.690	1011
2225	TAPERED	229.19 C	0.18	-780.99	1.05
2226	TAPERED	PASS	BS-4.7 (C)	0.692	1011
2227	TAPERED	240.79 C	-0.16	780.99	0.00
2228	TAPERED	PASS	BS-4.7 (C)	0.642	1012
2229	TAPERED	211.77 C	0.26	-791.03	0.00
2230	TAPERED	PASS	BS-4.3.6	0.601	1112
2231	TAPERED	255.91 C	0.06	687.94	0.00
2232	TAPERED	PASS	BS-4.7 (C)	0.516	1013
2233	TAPERED	483.76 C	0.40	-604.07	0.00
2234	TAPERED	PASS	BS-4.7 (C)	0.522	1013
2235	TAPERED	532.35 C	0.76	-572.05	0.00
2236	TAPERED	PASS	BS-4.7 (C)	0.454	1121
2237	TAPERED	463.58 C	33.90	-179.24	0.00
2238	TAPERED	PASS	BS-4.7 (C)	0.727	1121
2239	TAPERED	493.08 C	-77.34	160.68	3.22
2240	TAPERED	PASS	BS-4.7 (C)	0.414	1121
2241	TAPERED	406.11 C	-36.82	-146.14	0.00
2242	TAPERED	PASS	BS-4.7 (C)	0.485	1121
2243	TAPERED	402.72 C	-36.84	263.08	0.57
2244	TAPERED	PASS	BS-4.7 (C)	0.702	1121
2245	TAPERED	329.38 C	-61.67	371.45	2.27
2246	TAPERED	PASS	BS-4.7 (C)	0.497	1121
2247	TAPERED	221.21 C	-27.27	-375.28	0.00
2248	TAPERED	PASS	BS-4.7 (C)	0.509	1013
2249	TAPERED	260.43 C	0.04	-553.67	1.05
2250	TAPERED	PASS	BS-4.7 (C)	0.504	1001
2251	TAPERED	244.19 C	-0.07	551.00	0.00
2252	TAPERED	PASS	BS-4.7 (C)	0.480	1122
2253	TAPERED	146.61 C	-0.11	498.35	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	METE (UNLESS OTHERWISE NOTED)	RATIO/ MZ	LOADING/ LOCATION
2309	TAPERED	PASS 61.36 C	BS-4.7 (C) 0.02		0.380	1122
2310	TAPERED	PASS 29.04 C	BS-4.7 (C) -0.29		370.77 0.378 278.51	0.00 1022 1.37
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L UNITS ARE - KN						
2311	TAPERED	PASS 16.63 C	BS-4.7 (C) -0.89		0.547	1013
2312	TAPERED	PASS 58.46 C	BS-4.3.6 0.71		484.47 0.540	1.50 1011
2313	TAPERED	PASS 10.09 C	BS-4.3.6 -0.45		370.38 0.545	0.28 1013
2314	TAPERED	PASS 5.39 C	BS-4.7 (C) 28.33		616.37 0.520	0.00 1121
2315	TAPERED	PASS 5.41 C	BS-4.7 (C) -28.33		145.31 0.534	1.50 1121
2316	TAPERED	PASS 100.52 C	BS-4.7 (C) -0.03		-145.31 0.448	0.00 1011
2317	TAPERED	PASS 167.37 C	BS-4.7 (C) 0.12		-431.41 0.556	1.05 1011
2318	TAPERED	PASS 209.58 C	BS-4.7 (C) 0.26		-578.91 0.578	1.05 1011
2319	TAPERED	PASS 219.08 C	BS-4.7 (C) -0.23		-648.69 0.580	1.05 1011
2320	TAPERED	PASS 192.23 C	BS-4.7 (C) 0.38		648.69 0.535	0.00 1012
2321	TAPERED	PASS 330.17 C	BS-4.7 (C) 51.72		-653.81 0.638	0.00 1121
2322	TAPERED	PASS 452.28 C	BS-4.7 (C) -4.91		-367.13 0.449	0.00 1013
2323	TAPERED	PASS 496.24 C	BS-4.7 (C) -1.80		-479.03 0.426	0.00 1013
2324	TAPERED	PASS 513.72 C	BS-4.7 (C) 32.54		-442.40 0.433	0.00 1121
					-141.40	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

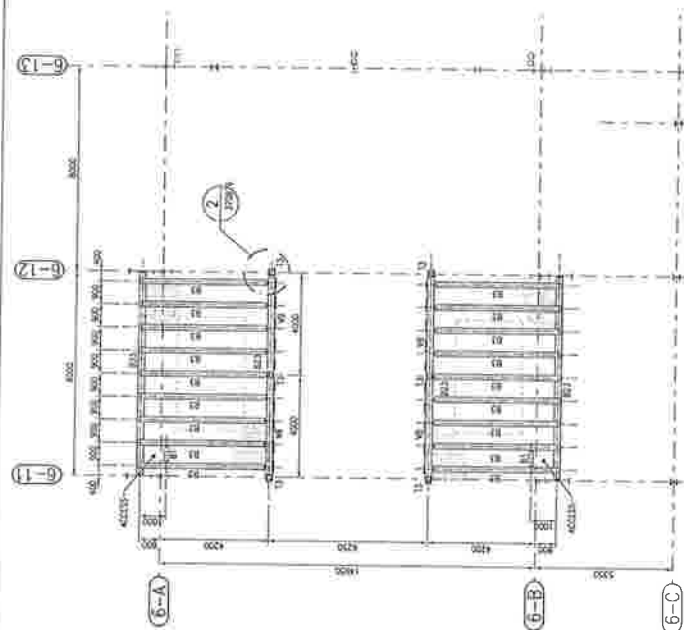
Appendix B

Design Drawings – Layout Plan

THIS DRAWING IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCHITECTS, M&E ENGINEER'S AND OTHER APPROPRIATE THIRD-PARTY DRAWINGS AND PERFORMANCE AND DESIGN SPECIFICATIONS.

1. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DWG 370401 TO 370404 FOR GENERAL NOTES AND TYPICAL DETAILS.

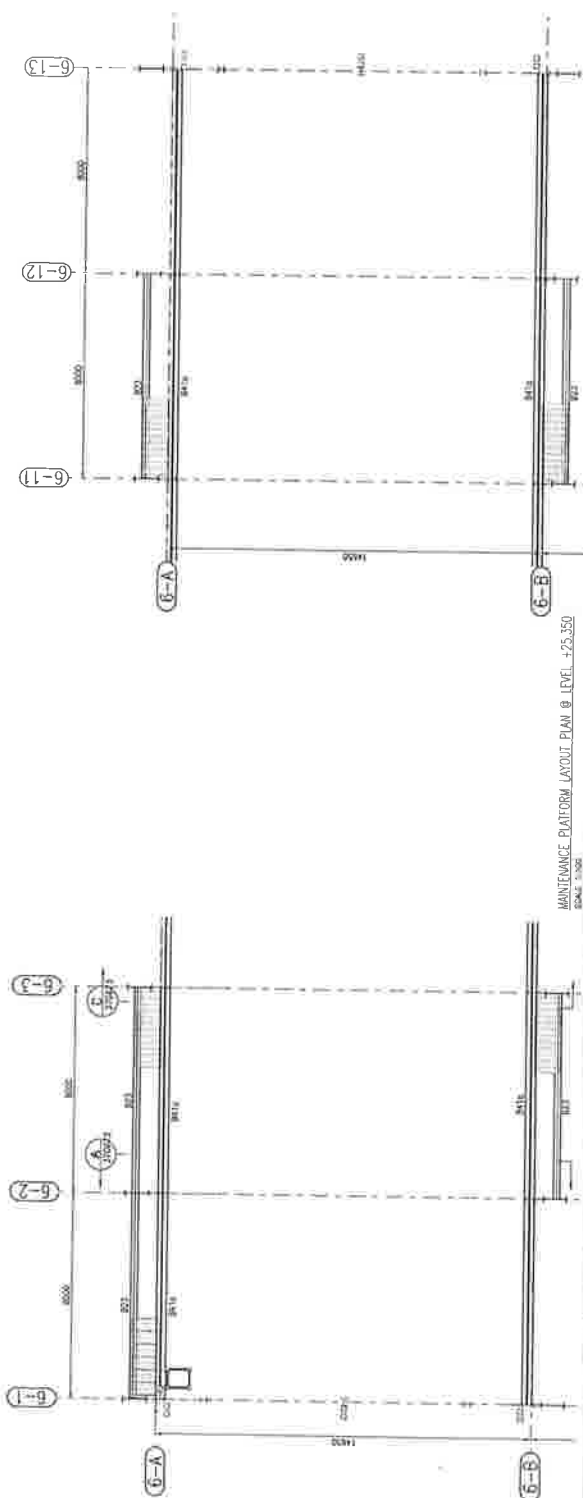
STEEL MEMBER SCHEDULE		
SECTION	SIZE	WEIGHT
10	10" x 10" x 1/2"	112.0
12	12" x 12" x 1/2"	153.0
14	14" x 14" x 1/2"	194.0
16	16" x 16" x 1/2"	235.0
18	18" x 18" x 1/2"	276.0
20	20" x 20" x 1/2"	317.0
22	22" x 22" x 1/2"	358.0
24	24" x 24" x 1/2"	399.0
26	26" x 26" x 1/2"	440.0
28	28" x 28" x 1/2"	481.0
30	30" x 30" x 1/2"	522.0
32	32" x 32" x 1/2"	563.0
34	34" x 34" x 1/2"	604.0
36	36" x 36" x 1/2"	645.0
38	38" x 38" x 1/2"	686.0
40	40" x 40" x 1/2"	727.0
42	42" x 42" x 1/2"	768.0
44	44" x 44" x 1/2"	809.0
46	46" x 46" x 1/2"	850.0
48	48" x 48" x 1/2"	891.0
50	50" x 50" x 1/2"	932.0
52	52" x 52" x 1/2"	973.0
54	54" x 54" x 1/2"	1014.0
56	56" x 56" x 1/2"	1055.0
58	58" x 58" x 1/2"	1096.0
60	60" x 60" x 1/2"	1137.0
62	62" x 62" x 1/2"	1178.0
64	64" x 64" x 1/2"	1219.0
66	66" x 66" x 1/2"	1260.0
68	68" x 68" x 1/2"	1301.0
70	70" x 70" x 1/2"	1342.0
72	72" x 72" x 1/2"	1383.0
74	74" x 74" x 1/2"	1424.0
76	76" x 76" x 1/2"	1465.0
78	78" x 78" x 1/2"	1506.0
80	80" x 80" x 1/2"	1547.0
82	82" x 82" x 1/2"	1588.0
84	84" x 84" x 1/2"	1629.0
86	86" x 86" x 1/2"	1670.0
88	88" x 88" x 1/2"	1711.0
90	90" x 90" x 1/2"	1752.0
92	92" x 92" x 1/2"	1793.0
94	94" x 94" x 1/2"	1834.0
96	96" x 96" x 1/2"	1875.0
98	98" x 98" x 1/2"	1916.0
100	100" x 100" x 1/2"	1957.0



LAND PLANT FROM LAROCHE PINN. 01151 + 27.558

IMPOSED LOAD : -1 kN/m^2 (MAINTENANCE PERSONNEL)

- 1 SKN/m² (MAINTENANCE PERSONNEL)
- 1 UNIT OF AHU WITH ACCESSORIES @ 30K



MAINTENANCE PLATFORM LAYOUT PLAN @ LEVEL +25.350

[illegible]